Informatica® Cloud Data Governance and Catalog

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Data Profiling for Assets

Informatica Cloud Data Governance and Catalog Data Profiling for Assets
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Preface

Read *Data Profiling for Assets* to learn how to create and run data profiles on data elements and view detailed profile results.

CHAPTER 1

Introduction to data profiling

You can determine the quality of data across sources and understand the completeness, conformity, and consistency of data in your organization with the help of data profiling.

Use data profiling to determine the characteristics of column data in source objects such as tables and views. When you run a profiling task on data elements, data profiling analyzes source data and displays column level statistics. You can see statistics such as value distribution, patterns, and data types in the profile results. These statistics help you discover the content and structure of data and determine the suitability of data to solve business problems.

When you run a profiling task, the results include the following column statistics:

- Number of distinct, non-distinct, and null values
- Percentage of distinct, non-distinct, null, and blank values
- Documented and inferred data types
- Number of patterns
- Percentage of top pattern
- Maximum and minimum length of values
- Maximum and minimum values
- Value frequencies and outliers

After you run a profiling task, you can perform the following actions:

- View historical and latest profile results.
- Compare two profile runs to analyze the statistics.
- Drill down on values, data types, or patterns.
- Export profile results to a Microsoft Excel file.

For more information about data profiling statistics, see *Understanding Technical Assets* in the Data Governance and Catalog help.

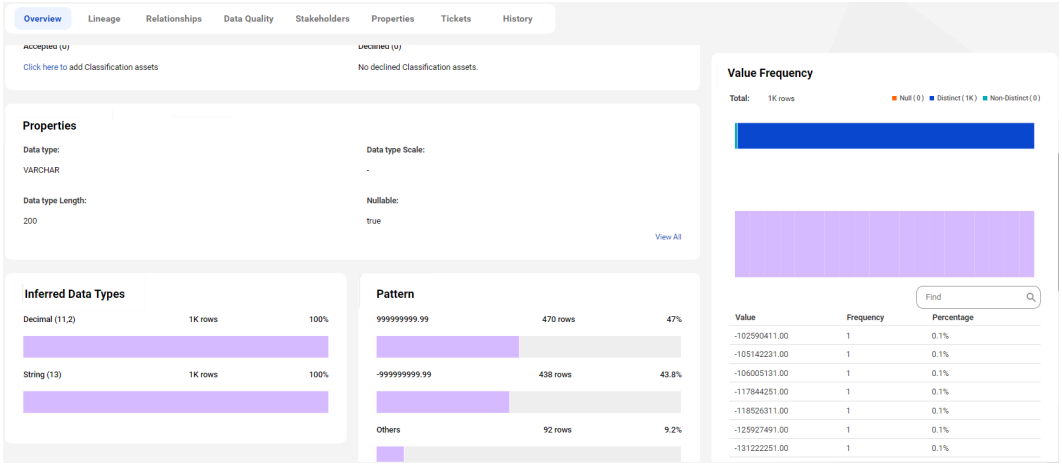
Data profiling process flow

You can enable data profiling after you create and run a catalog source in Metadata Command Center to extract metadata from the source system. Then choose to enable the data profiling capability for the catalog source and configure the data profiling parameters.

After you run data profiling as part of the catalog source job, the scope of profiling statistics that appear in Data Governance and Catalog depends on the data profiling parameters that you set when you configured the

catalog source. For each column, you can view statistics such as the characteristics, value frequencies, inferred data types, and patterns.

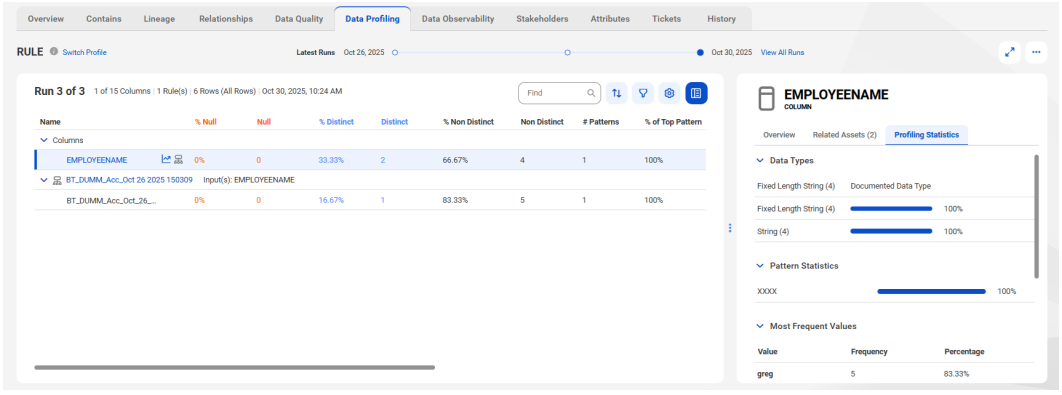
The following image shows the data profiling statistics that appear for a column in the **Overview** tab in Data Governance and Catalog:



If you associate one or more data quality assets as rules with the columns, you can retrieve more detailed profiling results.

To view the advanced column statistics for a data set, go to the **Data Profiling** tab and create a data profile for the data set in Data Governance and Catalog. You can add columns and data quality assets, such as rule specification, cleanse, parse, and verifier assets as rules to the profile. After you add columns as rules to the profile, you can run the profile to view detailed profiling statistics for each column associated with the profile. The profiling results are calculated based on the associated rules.

The following image shows the data profiling statistics that appear for a data set in the Data Profiling tab in Data Governance and Catalog:



You can define data profiling operations in Data Governance and Catalog and in the Data Profiling service in Intelligent Data Management Cloud™ (IDMC). The service that you choose for data profiling depends on the requirements of your data project and your organization. You can add data quality assets to your profiling jobs in both services.

Preparing for data profiling

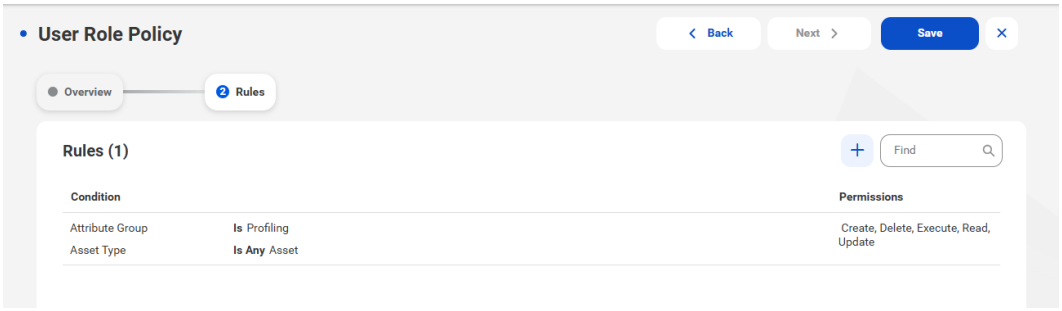
Before you create and run data profiles, ensure that you complete the prerequisites and enable data profiling in Metadata Command Center.

Complete the following prerequisites:

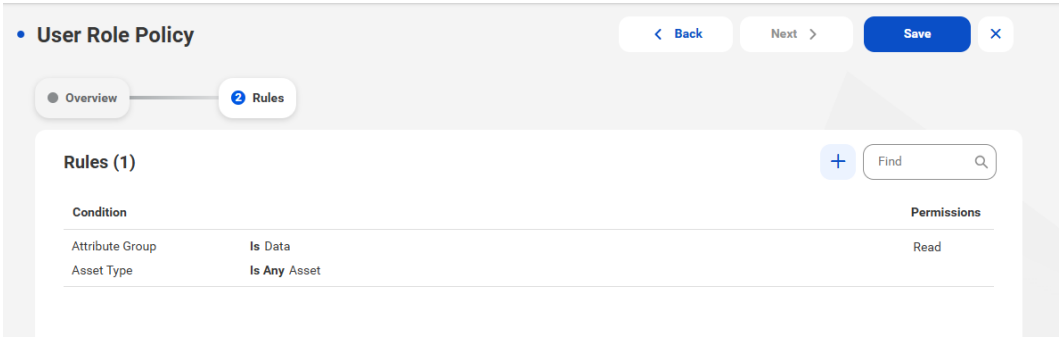
- In Administrator, create a user role and provide the required asset and feature privileges.
- Verify that the administrator of your organization configured a user role policy for your user role with the following conditions:
  - Create, Read, Update, Delete, and Execute permissions for the Profiling attribute group. You need these permissions to manage data profiles.
  - Read permission for the Data attribute group. You need these permissions to view value frequencies for data elements after you run a data profile and to preview failed rows for data quality rule occurrences.

  Note: If you create a user group policy that applies to all users in your organization, ensure that the policy grants permissions to create and run data profiles.
- If you assign another user as a stakeholder to the asset, verify that the administrator has configured a stakeholder policy for the user role.
- To view data quality assets, ensure that you have Read permission on the assets.
- To view data classifications, ensure that your organization administrator granted the View Data Classifications feature privilege to your user role in Administrator.

The following image shows a user role policy with the Create, Read, Update, Delete, and Execute permissions for the Profiling attribute group:



The following image shows a user role policy with the Read permission for the Data attribute group:



Perform the following tasks in Metadata Command Center:

- Create a catalog source and configure data profiling for the catalog source.

  Note: The parameters that you configure in Metadata Command Center also apply to data profiling in Data Governance and Catalog with the exception of filters and sampling type.

- Run a catalog source job to extract metadata from data sets.

Note: When you compare profile results over time, you can only see rules that are associated with columns on which you have access permissions.

Data profiling components

When you create a profile, you specify data elements such as columns and associate one or more data quality assets as rules with the data elements. Data profiling extracts column statistics from the data elements that you select.

Working with data elements

The following table describes the column precision ranges that apply to the data elements that you specify:

Data Type	Column Precision Range
Numeric	0 through 38
Non-numeric	0 through 4000

The following table describes the properties that you can view for the data elements that you specify:

Property	Description
Name	The name of the data element.
Business Name	The business name of the data element.
Data Type	The data type of the data element.
Precision	The number of characters in the data element.
Scale	The number of numeric characters after the decimal point.
Nullable	Indicates whether the data element can accommodate a NULL value.
Glossaries	Glossary assets that are associated with the data element.
Classifications	Data classifications associated with the data element.

To sort the list of columns in ascending or descending order, click the property name or the **Sort** icon. Use the **Find** box to search for columns.

Working with rules

You can add one or more data quality assets as rules. Select a rule specification, cleanse, parse, or verifier asset to associate with data elements.

For an asset that has a single input field, associate one or more columns for the field. For an asset that has multiple input fields, choose a column for each field. You can also automatically assign rules to columns based on column name and field name match.

CHAPTER 2

Data profiling operations

You can create, clone, modify, or delete data profiles.

To perform data profiling operations, open a data set and click the **Data Profiling** tab.

You can create multiple profiles for a data set. To switch data profiles, click **Switch Profile**.

Creating a profile

Create a profile and associate data elements with data quality assets as rules. When you create a profile, you choose a sampling type and apply filters to the profile.

1. Open the data set for which you want to create a profile.
2. On the **Data Profiling** tab, perform one of the following actions:
   - To create a profile for a data set for the first time, click **Create Profile**.
   - To create an additional profile for a data set, click the action menu and select **Create Profile** > **New Profile**.

   The **New Data Profile** page appears.
3. In the **General Information** area, enter a name and an optional description for the profile.
4. In the **Profile Settings** area, enter the following properties for the data profile:

Property	Description
Sampling Type	Defines the set of rows on which data profiling runs. Choose one of the following options: - All Rows. Runs data profiling on all rows. - Limit N Rows. Runs data profiling on a limited number of rows. - Custom Query. Uses an SQL clause to select the rows on which data profiling runs.
No of Rows to Limit	Required if you select Limit N Rows as the sampling type. Specifies the number of rows on which data profiling runs.
Sampling Query	Required if you select Custom Query as the sampling type. Enter a custom SQL query to define the set of rows on which data profiling runs. For example: `WHERE LAST_NAME LIKE 'P%'`

5. Optionally, in the **Filters** area, select **Yes** to define one or more filters that the profile can apply to the data.

You can define filter conditions as follows:

a. Choose a data element and an operator, and then enter one or more filter values. Multiple values within the filter have an OR relationship.

b. Click **Add** to define an additional filter. The filters you define have an AND relationship.

The following image shows three filters:



6. In the **Data Elements** area, click **Add** to choose the data elements that you want to include in the profile.

   Note: You can choose up to 50 data elements.

7. Click **Next**.
8. Add one or more rules to the profile.
9. Click **Save** to save the profile, or click **Save and Run** to save the configuration and run the profiling job.

Adding rules to a profile

Associate data quality assets as rules to data elements.

1. After you configure the properties on the **Overview** tab, click **Next** to go to the **Rules** tab.
2. On the **Rules** tab, click **Add Rule**.

   The **Add Rule** page appears.

3. Choose the data quality asset that you want to run as a rule within the profile, and click **Next**.

   You can choose a rule specification, cleanse, parse, or verifier asset.

4. Select the data elements to which you want to apply the rule.

   If the data quality asset has a single input field, choose one or more columns for the rule field.

   If the asset has multiple input fields, choose a single column for each rule field.

   Note: To automatically associate rule fields with columns based on matches between the rule field and the column names, click **Auto-assign**.

5. Optionally, add additional rules.
6. Click **Finish**.

Editing a profile

Edit a profile that you created to change the name, description, sampling type, filters, and rule associations.

1. Open the data set for which you want to edit a profile.
2. If multiple profiles exist for the data set, click **Switch Profile** and select the profile that you want to edit.
3. On the **Data Profiling** tab, perform one of the following actions:

- To edit a profile before it runs for the first time, click **Edit Profile**.
- To edit a profile that has associated profiling results, click the action menu and select **Edit Profile**.

4. Modify the required properties and rules, and click **Next**.
5. Click **Save** to save the profile, or click **Save and Run** to save the configuration and run the profiling job.

Cloning a profile

Clone a profile to create a copy of the profile. You can then modify the configuration of the profile that you cloned based on your requirements.

1. Open the data set for which you want to clone a profile.
2. If multiple profiles exist for the data set, click **Switch Profile** and select the profile that you want to clone.
3. On the **Data Profiling** tab, click the action menu and select **Create Profile** > **Clone Profile**.

   After you clone the profile, you can update the profile configuration and the rules that the profile uses.
4. Click **Save** to save the profile, or click **Save and Run** to save the configuration and run the profiling job.

Running a profile

Run a profile to discover a range of statistical information about a data set. Data profiling creates a job when you run a profile.

1. Open the data set for which you want to run a profile.
2. If multiple profiles exist for the data set, click **Switch Profile** and select the profile that you want to run.
3. In the **Data Profiling** tab, perform one of the following actions:
   - To run a profile for the first time, click **Run**.
   - To run a profile that has associated profiling results, click the action menu and select **Run Now**.

   A job is triggered to run the profile. You can monitor the status of the job and download logs on the **Job Monitoring** page.

Deleting a profile

Delete a profile to remove the profiling configuration and results associated with any job that ran for the profile.

1. Open the data set from which you want to delete a profile.
2. If multiple profiles exist for the data set, click **Switch Profile** and select the profile that you want to delete.
3. In the **Data Profiling** tab, click the action menu on the top of the page and select **Delete Profile**.

   A confirmation box appears.

4. Click **Delete**.

   The profile is deleted.

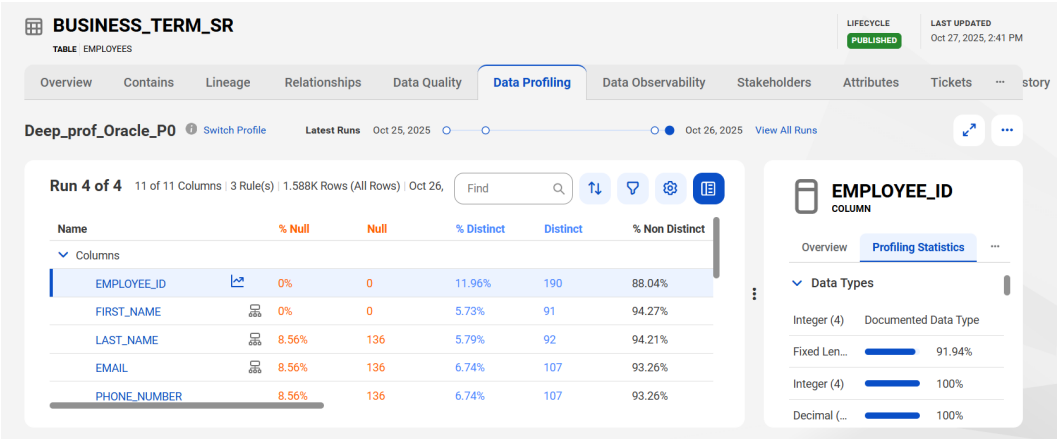
CHAPTER 3

Viewing profiling results

You can view the results for a profiling job after you run the profiling task. The profile results include the characteristics of columns such as value frequency, patterns, and data types.

The page with the profile results contains a header area that includes the profile name and a scale that indicates the last ten job instances. To view a summary for a job instance, hover over the point corresponding to the job instance.

The following image shows the results of data profiling on a data set:



You can view the following information:

- Run number
- Number of columns and rules associated with the profile
- Number of profiled rows and the row sampling type
- The date and time on which the profile ran

Use the sort option to sort the column statistics in ascending or descending order. You can search for results or apply filters to see relevant results. To add or remove a column statistic, right-click the name and then select or clear the column statistic.

Profile statistics

The profile statistics for each column appear in the profile results table. The table also displays rule statistics for the output values obtained from each rule.

The column statistics appear in a collapsible section. You can hover over the Rule icon to view the name of the rule associated with the column.

Note: The icon changes when columns have multiple rule associations.

The following table describes the statistics that you can view after you run a profile:

Property	Description
Null	The number of null values in the column or the rule output.
% Null	The percentage of null values in the column or the rule output.
% Distinct	The percentage of distinct values in the column or the rule output.
Distinct	The number of distinct values in the column or the rule output.
% Non-distinct	The percentage of non-distinct values in the column or the rule output.
Non-distinct	The number of non-distinct values in the column or the rule output.
# Patterns	The number of patterns in the column or the rule output.
% of Top Pattern	The percentage of values with the most frequent pattern in the column or the rule output.
Maximum Length	The length of the longest value in the column or the rule output.
Minimum Length	The length of the shortest value in the column or the rule output.
Minimum Value	The lowest value in the column or the rule output.
Maximum Value	The highest value in the column or the rule output.
Blank	The number of empty values in the column or the rule output.
% Blank	The percentage of empty values in the column or the rule output.

When you select a column, a detailed view of the profile results for the column data appears in the Preview pane. The detailed view includes statistics such as data types, pattern statistics, numeric column statistics, most frequent values, and trends.

Note: You can add or remove associated rules using the action menu. Also, you can use the action menu to go to the **Job Monitoring** page to view profiling jobs.

After you run a profile, you can perform the following tasks:

- Drill down on the profile results for data that you select. Select a value, data type, or pattern to view relevant data in the Data Preview area.
- Export a part of the profile results or the complete results to a Microsoft Excel file.

  Note: You can choose to include data from one or more columns or rules.

Viewing results for a profiling job instance

You can view the most recent results for a profile job instance or the results from any of the last ten occasions on which the profile job ran.

1. Open the data set for which you want to view profile results.

2. If multiple profiles exist for the data set, click **Switch Profile** and select the profile for which you want to view the results.
3. To view results from a profiling job instance, perform one of the following actions:
   - From the scale that displays the last ten profiling job instances, click the point corresponding to the profiling job instance that you want to view.
   - Click **View All Runs**, choose the profiling job instance that you want to view, and click **Select**.

   The results for the profiling job instance that you selected appear.

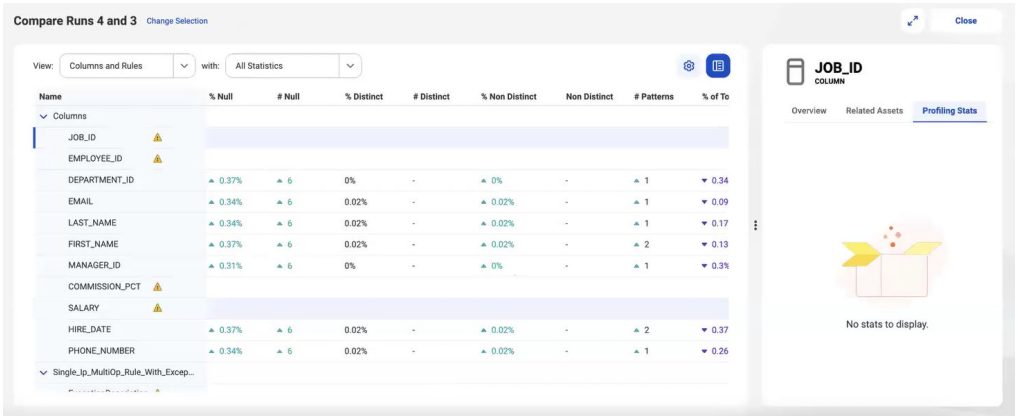
Comparing profile results over time

You can compare profile results from two different job instances to observe the changes in your source data over time.

1. Open the data set for which you want to compare profile results over time.
2. If multiple profiles exist for the data set, click **Switch Profile** and select the profile for which you want to compare the results.
3. On the **Data Profiling** tab, click the action menu on the top of the page and select **Compare** > **Profile Runs**.

   The **Compare Profile Runs** window appears.
4. Choose any two job instances to compare, and click **Select**.

   The comparison of column statistics from both runs appear. An up arrow with a numeric count indicates an increase in the statistic compared to the earlier run. A down arrow with a numeric count indicates a decrease in the statistic. A warning icon besides a column name indicates that the column was not a part of one of the profile instances. You can hover over the icon to see the details.
   The following image shows a comparison of column statistics:



5. Optionally, customize the view based on your requirements.

   You can sort the compared results in ascending or descending order. You can apply filters or search for a result. To add or remove a column statistic, right-click the name and then select or clear the column statistic.

   When you click a column, a detailed view of the changes for the column appears in the preview pane. The profile results include statistics such as data types, patterns, and values.

The following table describes the options that you can use to filter the data:

Option	Description
View	Displays data for columns, rules, or for both columns and rules.
With	Filters the results based on the type of comparison. You can choose from the following types: - All Statistics. View all results for both the runs. - Differences. View the results that are different between the runs. - Matches. View the results that match in both runs. - Added. View the results for columns added in the later run. - Removed. View the results for columns removed in the later run.
Find	Looks for results that contain a keyword that you enter.

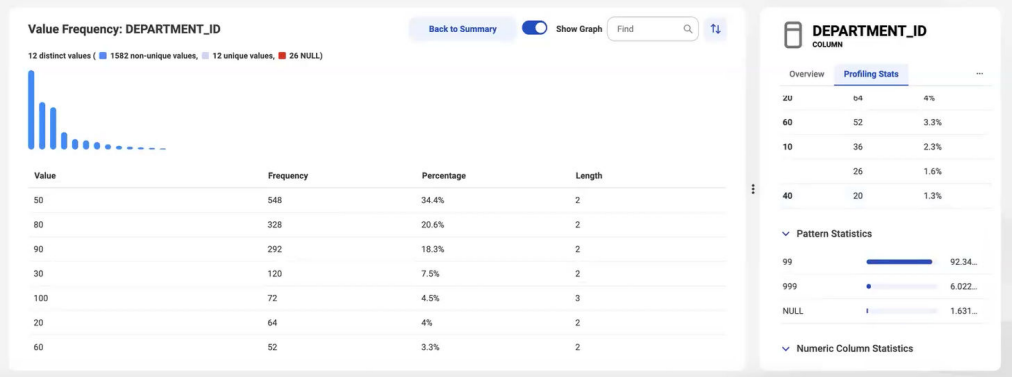
6. Click **Close** to return to the results page.

Viewing value frequencies

View the distribution of null values, unique values, and non-unique values in a bar chart for a column or rule. You can view value frequencies for each column in the summary view and detailed view.

1. Open the data set for which you want to view value frequencies.
2. If multiple profiles exist for the data set, click **Switch Profile** and select the profile from which you want to view value frequencies.
3. On the **Data Profiling** tab, choose the column or rule for which you want to view value frequencies.
4. Click the Graph icon to open the detailed view.

   The following image shows the value frequencies for a sample column:



   To hide the bar chart, clear the **Show Graph** option. Click **Back to Summary** to return to the summary view.

Drill down on the profile results

Drill down on column values, rule output values, data types, or patterns in the results of a profiling task.

1. Open the profile results on which you want to drill down.
2. Choose the column or rule on which you want to drill down.
3. Click the Graph icon to open the detailed view.
4. Select the values from the **Value Frequency** area or select the data types and patterns from the preview pane.

   The values, data types, or patterns that you select are added as filters.
5. Expand the **Data Preview** panel and click **Apply Filters**.

   A job is triggered to drill down on the profile results. You can monitor the status of the job on the **Job Monitoring** page.
6. After the job completes, view the results in the **Data Preview** panel.

   Note: You can view results from the top 100 rows. Click **Download** to download the results as a .csv file.

Export profiling results

Export the profiling results for one or more columns or rules to a Microsoft Excel file.

1. Open the profile results that you want to export.
2. If multiple profiles exist for the data set, click **Switch Profile** and select the profile that you want to export.
3. Click the action menu and click **Export Profiling Results**.
4. In the **Export Profiling Results** window, configure the following properties:

Property	Description
File Name	The name of the Microsoft Excel file that the export operation creates. You can retain the default file name or specify a file name of your choice.
Range	Specifies whether the export file includes data from all columns or from one or more columns that you specify.
Columns	Required if you choose **Selected Columns** in the Range property. Identifies the columns or rules to include in the profiling results.
Scope	Specifies whether to include summary, value frequencies, statistics, patterns, data types, or all the profile results. Choose one or more of the following options: - Summary. Exports all the profile results. - Value Frequencies. Exports the value frequencies. - Statistics. Exports the profiling statistics. - Patterns. Exports the patterns. - Data Types. Exports the documented and inferred data types.

5. Click **Export**.

A job exports the profile results. You can monitor the status of the job on the **Job Monitoring** page.

6. After the job completes, click **Download Export File** to download the profiling results as a Microsoft Excel file.